LAB EXPERIMENT
ITA0506-COMPUTER VISION

PAVENDHAN A -192321109

EXPNO:22

Aim

To sharpen an image using a Laplacian mask with a positive center coefficient.

Laplacian Mask

A commonly used 4-neighbor Laplacian mask with a positive center coefficient is:

0 -1 0

-1 4 -1

0 -1 0

For sharpening, the Laplacian result is added to the original image:

Sharpened Image = Original Image + Laplacian Image

Algorithm

1. Read the input image.
2. Convert the image to grayscale.
3. Define the Laplacian mask with a positive center coefficient.
4. Apply the mask using convolution.
5. Add the Laplacian image to the original image.
6. Limit pixel values between 0 and 255.
7. Display the original and sharpened images.

Code:

EXP-22.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...

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```python
import cv2
import numpy as np

# Step 1: Read the image
image = cv2.imread("BEN 10.png")

# Check whether image is loaded
if image is None:
    print("Error: Image not found!")
    print("Make sure BEN 10.png is in the same folder as this Python file.")
    exit()

# Step 2: Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Step 3: Define Laplacian mask
# Positive center coefficient
mask = np.array([
    [0, -1,  0],
    [-1, 4, -1],
    [0, -1,  0]
], dtype=np.float32)

# Step 4: Apply Laplacian mask
laplacian = cv2.filter2D(gray, cv2.CV_32F, mask)

# Step 5: Sharpen the image
sharpened = gray.astype(np.float32) + laplacian

# Step 6: Keep pixel values between 0 and 255
sharpened = np.clip(sharpened, 0, 255)

# Convert to 8-bit image
sharpened = sharpened.astype(np.uint8)

# Step 7: Convert Laplacian result for display
laplacian_display = cv2.convertScaleAbs(laplacian)

# Step 8: Display images
cv2.imshow("Original Image", gray)
```

Ln: 10 Col: 27

OUTPUT :

Result

The image is successfully sharpened using a Laplacian mask with a positive center coefficient. The edges and fine details of the image become more prominent.